

Navier–Stokes: Recovering Minimal Structure from Representation-Induced Complexity

Michael Zot

Independent Researcher

Corresponding author: mike@ZotBot.Ai

Research repository: github.com/mikecreation/ZotBot/tree/main/research/navier-stokes-spc

Abstract

A proof contains both mathematical dependence and the history of one successful route through a particular representation. These structures can have very different complexity. We develop *Structural Path Compression*, a method for recovering a smaller generative architecture from a successful reasoning trajectory by separating same-grade dependency from higher-grade feedback. The framework is motivated by the author’s earlier Minimal Irreversible Predictive Update (MIPU) and Maximal Anticipatory Pattern (MAP) formalism [2].

We apply the method to the residual-improvement construction in the 2026 OpenAI Navier–Stokes blowup preprint [1]. The published Section 9 construction advances a synchronized stage parameter $\sigma_j = 1/5 + j/10$ through a repeated correction cycle. Reorganizing the same local correction primitives about the fixed initialized state yields an admissible filtered error system. Its proposed stage-free reduction has two load-bearing estimates:

$$K\mathfrak{F}_{\text{adm}}^s \subseteq \mathfrak{F}_{\text{adm}}^{s+\delta_L}, \quad \delta_L = \frac{1}{2} - 4\kappa_s > 0,$$

$$Q_{\text{ret}}(\mathcal{X}^s, \mathcal{X}') \subseteq \mathfrak{F}_{\text{adm}}^{s+t-2\kappa_s}.$$

For $\kappa_s = 10^{-5}$, the linear return margin is 0.49996, while the corresponding nonlinear difference gain from seed grade $\sigma_0 = 1/5$ is 0.19998. Once these estimates are established, arbitrary finite residual order follows from a finite Neumann polynomial and a finite filtered Picard calculation. The existing common-domain, physical-derivative, and shrinking-cutoff arguments of the source construction can then be read through an accuracy-indexed rather than stage-indexed interface.

The mathematical claim is therefore narrower than a wholesale shortening of the 166-page source proof: the proposed compression targets the repeated residual-improvement architecture of Section 9. The broader result is methodological. A successful proof path can contain obligations created by its representation; a change of representation can delete those obligations while preserving the theorem and downstream interface. The manuscript states explicit failure criteria so that the compression can be independently audited.

Keywords. Navier–Stokes; proof architecture; representation change; proof compression; filtered algebra; theorem proving; MIPU; MAP.

MSC 2020. 35Q30; 68T15; 68T20.

Source under analysis. OpenAI, *Finite Time Blowup for Navier–Stokes*, 2026, especially Sections 7–9 [1].

Main result at a glance

Structural claim. The repeated +0.1 stage schedule in Proposition 9.6 of the source construction is proposed to be replaceable, at its downstream realization interface, by a fixed-background filtered compiler once two positive return estimates are verified. The local Section 7–8 correction machinery is retained; the repeated chronological schedule is the object being removed.

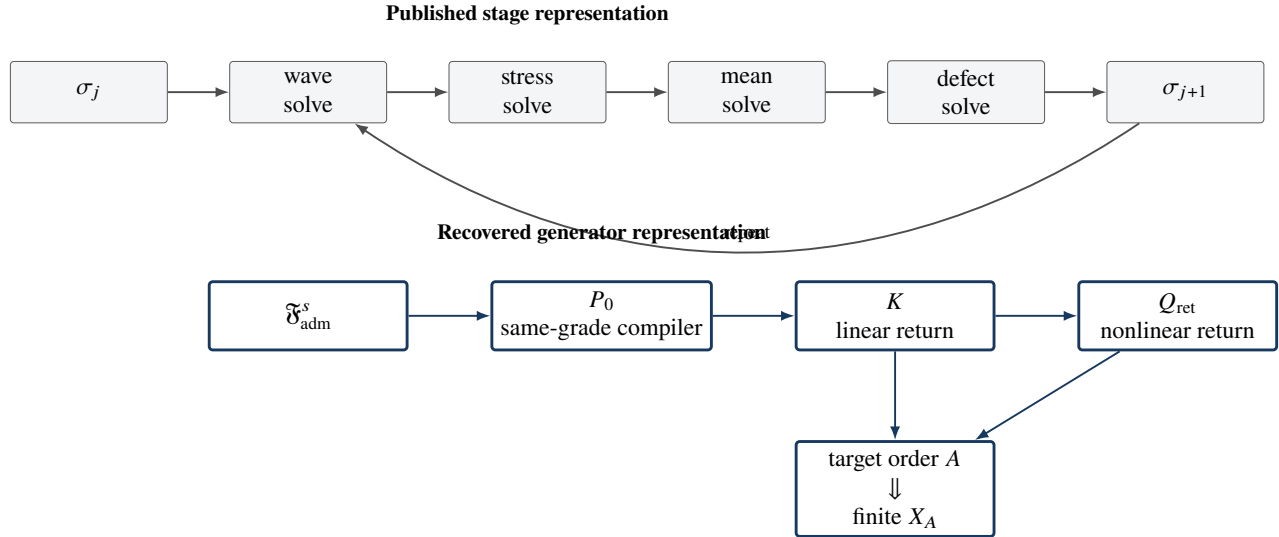


Figure 1. The proposed representation change. The source construction synchronizes several unequal gains through a repeated +0.1 stage clock. The compressed representation attempts to expose the same-grade finite dependency graph and treat all returns by their positive filtration degree.

Proposition 0.1 (Two load-bearing estimates). *Let U_0 denote the initialized state of Proposition 9.5 of the source construction. If the legal one-pass compiler P_0 and the retained fixed-background error expansion satisfy*

$$K := I + L_{\text{ret}}P_0, \quad K\mathfrak{F}_{\text{adm}}^s \subseteq \mathfrak{F}_{\text{adm}}^{s+\frac{1}{2}-4\kappa_s},$$

and

$$Q_{\text{ret}}(\mathcal{X}^s, \mathcal{X}^t) \subseteq \mathfrak{F}_{\text{adm}}^{s+t-2\kappa_s},$$

then every finite target grade $A > 1/5$ admits a finite legal correction X_A whose retained error lies in $\mathfrak{F}_{\text{adm}}^A$, modulo flat pulse/cutoff remainders. The remaining passage to a smooth field uses the same common-domain, derivative-loss, and shrinking-cutoff interfaces already present in the source construction.

The paper is deliberately organized so that an independent reader can attack [proposition 0.1](#) without accepting the conceptual narrative surrounding it.

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1. The complexity of a theorem and the complexity of reaching it

A proof has at least two geometries: the geometry of mathematical dependence and the geometry of the route by which that dependence was discovered. A long successful trajectory establishes reachability; it does not by itself establish that every part of the trajectory belongs to the irreducible structure of the problem.

This distinction becomes sharper when mathematical discovery is computational. A machine can search a vast space of intermediate states, repeatedly repairing obligations created by one representation, before a later analysis recognizes that those obligations are generated by a much smaller rule. The resulting post-discovery question is

Which parts of a successful proof belong to the problem, and which belong to the representation?

The present paper studies that question in the residual-improvement machinery of the OpenAI Navier–Stokes construction [1]. The source paper is 166 pages; its Section 9 occupies printed pages 100–115 and coordinates local operators already developed in Sections 7–8. On printed page 106 the authors state that the base, primary amplitudes, and inverse operators are kept fixed throughout the correction construction; Proposition 9.6 then repeats the four-step cycle and advances all tracked orders by 0.1 [1, pp. 106–111]. Lemmas 9.7–9.8 provide the common physical domain and derivative-loss interface used by Proposition 9.9 [1, pp. 111–115].

The aim here is not to rewrite the complete Navier–Stokes proof. It is to identify whether the Section 9 stage schedule is an intrinsic mathematical grade or a synchronization grade introduced by the chosen proof organization.

1.1. Persistent information and inferential closure

Let D denote a mathematical state and R a representation. We write

$$\text{Cl}_R(D)$$

for the consequences accessible from D under the transformations made natural by R . Two representations can preserve the relevant mathematical content while exposing different amounts of immediate consequence:

$$R_1(D) \simeq R_2(D), \quad \text{Cl}_{R_1}(D) \subsetneq \text{Cl}_{R_2}(D).$$

Nothing new has been measured. The inferential closure has changed. Change-of-representation research in automated reasoning explicitly studies this phenomenon [6, 7]; proof planning similarly separates reusable high-level proof structure from low-level inference search [5].

Definition 1.1 (Representation-induced complexity). Representation-induced complexity is proof or search work that disappears under a faithful recoding while the hypotheses, conclusion, and downstream mathematical interface remain intact.

Remark 1.2. This definition is intentionally operational. It does not require a universal numerical measure of complexity. Deletion of an independent proof obligation is already evidence that some work belonged to the representation rather than the theorem’s invariant dependency structure.

2. MIPU, MAP, and Structural Path Compression

The conceptual language used here develops the author’s earlier MIPU/MAP framework [2]. A *Minimal Irreversible Predictive Update* (MIPU) is a smallest integrated update that changes what can be predicted or inferred next. A *Maximal Anticipatory Pattern* (MAP) is the widest coherent predictive structure supported by the information currently available.

For proofs, the translation is natural. A proof MIPU changes the future deductive state. A proof MAP is a stable dependency architecture that predicts an entire family of later steps.

Core law.

successful trajectory $\neq$ minimum sufficient trajectory.

A sequence of valid local updates can eventually reveal a generator that makes future instances of those updates predictable rather than independently proved.

2.1. Reprocess under recode

If a new representation R_2 enlarges the inferential closure of already possessed information D , then conclusions extracted under R_1 should not simply be carried forward unchanged. The original material should be revisited:

$$R_1 \rightarrow R_2 \implies \text{REPROCESS}(D).$$

A previously necessary-looking proof step may become a derived consequence. A repeated cycle may become a finite expansion of one transfer law. This operation is the bridge from MIPU/MAP to Structural Path Compression.

2.2. Horizontal and vertical structure

The central diagnostic is to separate *same-grade* dependence from *higher-grade* feedback. If the current-grade graph is acyclic, all obligations at that grade can be solved by finite forward substitution. If every return edge has strictly positive degree, repetition survives only as a move to higher filtration.

The target pattern is therefore

acyclic horizontally, increasing vertically.

This pattern is what the Navier–Stokes case appears to exhibit.

3. The source architecture under analysis

Section 9 of [1] assembles four correction mechanisms developed earlier in the paper:

1. a supported nonzero-harmonic pulse inverse (Proposition 7.2, printed p. 77);
2. a signed covariance/stress realization (Proposition 7.6, printed p. 84);
3. compactly supported mean corrections and exact compatibility identities (Section 8, especially Proposition 8.4 and Corollary 8.5, printed pp. 94–95);
4. a fixed five-equation compatibility correction (Lemmas 8.7–8.8, printed pp. 97–99).

Proposition 9.1 and Lemma 9.2 then quantify the residual and product orders created by these operations [1, pp. 101–103]. Proposition 9.3 explicitly permits any *finite sequence* of the named primitive corrections around the fixed slow base and primary waves, provided pressure is reconstructed from the complete current velocity and nonlinear products use the complete current field [1, pp. 103–105].

3.1. The published stage clock

Definition 9.4 uses

$$B_j = \frac{1}{2} + \sigma_j, \quad C_j^* = 1 + \sigma_j, \quad \sigma_j = \frac{1}{5} + \frac{j}{10}.$$

Proposition 9.6 performs the wave, averaged-stress, zero-average mean, and compatibility corrections, then shows that all tracked orders improve by 0.1. The closing page of the proposition explicitly takes the minimum of several larger gains and concludes that the wave, tangential mean, and defect orders all improve by 0.1 [1, p. 111].

This is the first structural clue: the 0.1 increment is a common synchronization rate, not the exact gain of the individual return channels.

4. Admissible filtration and recovered dependency graph

Freeze the initialized state $U_0 = (u^{[0]}, p^{[0]})$. The retained augmented error is organized into

$$W, \quad \bar{M}, \quad M^\circ, \quad S,$$

where W is the nonzero angular-harmonic residual, $\bar{M}$ is the auxiliary-averaged tangential mean residual, M° is its zero-auxiliary-average complement, and S stores the compact-support compatibility state (P, J_θ, J_z) .

Define

$$\mathfrak{F}^s = W_{\frac{1}{2}+s} \oplus \bar{M}_{1+s-\kappa_s} \oplus M^\circ_{1+s-2\kappa_s} \oplus S_{1+s-2\kappa_s}. \quad (4.1)$$

The actual compiler domain is the admissible subspace $\mathfrak{F}_{\text{adm}}^s$ satisfying the exact integrated identities inherited from Proposition 8.4:

$$\int R^2 \bar{E}_\theta \, dR = \varepsilon \partial_Z J_\theta, \quad \int R \bar{E}_z \, dR = \varepsilon \partial_Z (J_z + c_\rho P).$$

These relations are needed to prepare the compact stress target used by the signed-wave inverse.

4.1. The same-grade graph

At fixed grade the principal ordering is

$$\boxed{W \longrightarrow \bar{M} \longrightarrow M^\circ \longrightarrow S.}$$

There are shortcut arrows, but every same-grade dependency moves forward. The reverse arrows are higher-order returns.

The one-pass compiler P_0 is the finite forward substitution through this graph. Let

$$K := I + L_{\text{ret}} P_0.$$

The central linear statement is

$$K \mathfrak{F}_{\text{adm}}^s \subseteq \mathfrak{F}_{\text{adm}}^{s+\delta_L}, \quad \delta_L = \frac{1}{2} - 4\kappa_s. \quad (4.2)$$

For $\kappa_s = 10^{-5}$,

$$\delta_L = 0.49996.$$

The bottleneck is the return

$$\bar{M} \longrightarrow \text{signed-stress wave} \longrightarrow W,$$

whose exponent is already visible in the closing inequalities of Proposition 9.6 [1, p. 111].

5. The two estimates

The second load-bearing estimate concerns the physical quadratic map. The compiler uses typed provenance internally – in particular, the exact identity $B(W_0, L\Sigma) = \Sigma$ belongs only to waves in the image of Proposition 7.6, not to arbitrary members of the same norm class – but Q_{ret} acts on the realized physical correction fields.

Let the conservative correction envelope be denoted by $\mathcal{X}^s$. Then the proposed quadratic estimate is

$$Q_{\text{ret}}(\mathcal{X}^s, \mathcal{X}^t) \subseteq \mathfrak{F}_{\text{adm}}^{s+t-2\kappa_s}. \quad (5.1)$$

The worst channel is wave–wave interaction into the zero angular harmonic. Lemma 9.2 states that two waves of exponents α, α' produce a zero harmonic in $M_{\alpha+\alpha'-\kappa_s}$ [1, pp. 102–103]. With correction waves at the signed-stress envelope, this exactly saturates the averaged-mean threshold in (5.1).

Since the seed grade is $\sigma_0 = 1/5$, the nonlinear difference gain is

$$\eta = \sigma_0 - 2\kappa_s = 0.19998 > 0.$$

The resulting architecture is summarized by two positive return margins:

$$\delta_L = 0.49996, \quad \eta = 0.19998.$$

Table 1. Critical channels in the compressed architecture.

Type	Channel	Output role	Margin
Linear	$\bar{M} \rightarrow \text{signed-stress wave} \rightarrow W$	bottleneck backward return	$\frac{1}{2} - 4\kappa_s$
Quadratic	$W \times W \rightarrow \bar{M}$	bottleneck zero harmonic	seed gain $\frac{1}{5} - 2\kappa_s$
Linear	$M^\circ \rightarrow W$	higher-order backward return	$> \delta_L$
Linear	$S \rightarrow S$ after defect solve	higher-order defect remainder	$> \delta_L$

6. Finite algebra replaces infinite staging

Fix a target grade $A > \sigma_0$ and set

$$N_A := \min\{N : \sigma_0 + N\delta_L \geq A\}.$$

Define the finite polynomial

$$P_A = P_0 \sum_{n=0}^{N_A-1} K^n. \tag{6.1}$$

Because $L_{\text{ret}}P_0 = K - I$,

$$L_{\text{ret}}P_A = K^{N_A} - I.$$

Thus P_A is an inverse to the required finite resolution; no infinite Neumann series or new completion is required.

The nonlinear correction is computed by finite filtered Picard substitution:

$$Y_A^{(m+1)} = e_0 + Q_{\text{ret}}(P_A Y_A^{(m)}, P_A Y_A^{(m)}).$$

Successive differences gain at least $\eta > 0$, so after finitely many substitutions the iteration stabilizes modulo $\mathfrak{F}^A$. Setting $X_A = P_A Y_A$ yields

$$\mathcal{E}_{\text{full}}(U_0 + X_A) \in \mathfrak{F}_{\text{adm}}^A + \mathcal{I}_{\text{flat}}.$$

The key conceptual shift is therefore

$$\boxed{\text{stage number } j \longrightarrow \text{requested accuracy } A.}$$

Accuracy becomes an input to a finite generator rather than an inherited consequence of repeatedly advancing a clock.

7. Returning to physical space

The source proof’s Lemmas 9.7–9.8 are the last place where the deleted stage parameter could be carrying hidden analytic work. Lemma 9.7 gives one $q_{\text{big}} > 0$ independent of stage and derivative order; its proof attributes this uniformity to the fixed background, fixed primary covariance, fixed supports, and fixed inverse operators [1, pp. 111–113]. Lemma 9.8 gives physical derivative bounds with exponent losses independent of stage and only afterward inserts the specific stage fact that every retained increment has normalized exponent at least $j/10$ [1,

pp. 113–114].

The stage-free reading replaces that specific normalized exponent by the shell order A_{j-1} of an increasing accuracy sequence $A_j \rightarrow \infty$. Consequently one obtains correction orders and residual orders tending to infinity with fixed derivative losses, which is precisely the interface required by Lemma 5.4 and Proposition 9.9 [1, pp. 56–61, 114–115].

Theorem 7.1 (Stage-free replacement at the downstream interface). *Assume (4.2) and (5.1), together with legality of P_0 , preservation of admissibility, and the stage-free extraction of the common-domain and physical-derivative estimates from Lemmas 9.7–9.8 of [1]. Then the repeated law*

$$\sigma_{j+1} = \sigma_j + \frac{1}{10}$$

is unnecessary for producing the hypotheses consumed by the final shrinking-cutoff realization. Any sequence $A_j \rightarrow \infty$ generates finite legal states whose shell orders and physical residual orders tend to infinity with derivative losses independent of j .

8. What has actually been compressed

The local PDE machinery remains. The oscillatory inverse remains. The signed covariance realization remains. The mean inverse remains. The compatibility correction remains. Pressure reconstruction remains. Product estimates, support calculus, and the final shrinking-cutoff theorem remain.

What disappears is the need to use the particular repeated schedule

$$\sigma_j = \frac{1}{5} + \frac{j}{10}$$

as the primitive explanation of residual improvement.

Structural compression is measured by deleted obligations, not deleted pages. The proposed reduction does not turn a 166-page proof into three pages. It targets a specific repeated proof architecture inside Section 9. Its significance, if verified, is that infinitely many synchronized stage obligations become consequences of two reusable filtration estimates.

This distinction also clarifies why the source construction can be locally efficient and globally compressible at the same time. Proposition 9.6 already compresses several unequal gains into one safe +0.1 clock. Structural Path Compression asks whether that clock itself must be replayed once the return law has been isolated.

9. Representation-induced complexity and mathematical understanding

Classical proof complexity studies shortest proof length relative to formal systems [4]. Proof planning studies reusable high-level patterns that organize lower-level inference [5]. Automated change-of-representation work studies how equivalent encodings alter reasoning performance [6, 7]. Thurston’s discussion of mathematical progress emphasizes that understanding is not exhausted by possession of a formal derivation [3].

Structural Path Compression treats a successful proof as data for a second discovery problem:

What compact structure explains why this trajectory kept succeeding?

A long machine-generated proof can therefore be both a certificate and a measurement of the problem under one representation.

9.1. Discovery, structure, representation, verification

It is useful to distinguish four costs:

$$C_{\text{discovery}}, \quad C_{\text{structure}}, \quad C_{\text{representation}}, \quad C_{\text{verification}}.$$

Observed reasoning length is not a direct measurement of intrinsic structure. Expensive search can discover a small generator. Conversely, a concise presentation can hide substantial search or representation cost.

9.2. Machine-generated mathematics

Recent formal theorem-proving systems such as AlphaProof demonstrate the power of learned search in formally verified environments [8]. The natural post-solution extension is

$$\text{search} \rightarrow \text{verify} \rightarrow \text{extract dependencies} \rightarrow \text{search representations} \rightarrow \text{compress} \rightarrow \text{reprocess}.$$

The relevant long-run question becomes less “can the system reason far enough?” and more “can it discover why so much of its own reasoning was unnecessary?”

10. Falsification surface

A compression of a major proof should become easier to attack after reformulation, not harder. The present proposal fails if any of the following is established:

1. a zero-degree cycle exists in the same-grade dependency graph;
2. P_0 applies any primitive inverse outside its domain;
3. the signed covariance identity is used for an untyped generic wave rather than a wave in the image of Proposition 7.6;
4. Q_{ret} fails to preserve the admissibility identities;
5. an overlooked nonlinear channel removes the positive margin η ;
6. exact pressure reconstruction introduces algebraic degree greater than two around U_0 ;
7. a finite compiler output cannot be expanded into a legal finite sequence of the operations permitted by Proposition 9.3;
8. flat remainders can re-enter finite order under one of the allowed finite operations;
9. the common physical domain shrinks with target accuracy;
10. physical exponent losses grow with target order;
11. the downstream summation uses genealogical information specific to the original stage sequence;
12. higher target accuracy modifies lower-order shells, so $X_B - X_A \notin \mathcal{X}^A$ for some $B > A$.

Diagnostic 10.1 (Fastest route to disproof). The mathematically cheapest way to kill the proposed compression is to exhibit one legal same-grade feedback loop of total degree zero. The second-cheapest is to exhibit one quadratic channel whose loss is large enough to make $\eta \leq 0$.

11. Audit and reproducibility protocol

The manuscript is designed for independent checking at three levels.

11.1. Source fidelity audit

Every primitive operator invoked by the compiler should be cross-checked against the exact proposition or lemma in [1]. Appendix B provides the initial map.

11.2. Channel audit

Every arrow entering or leaving $W, \bar{M}, M^\circ, S$ should be assigned a filtration degree. Appendix C records the presently load-bearing channels and the claim each supports.

11.3. Execution audit

For several target grades A , expand P_A and the finite Picard expression into a literal sequence of source-legal primitive operations, then verify pressure reconstruction, support, admissibility, and the predicted shell order. This turns the abstract compiler into a reproducible proof object.

12. Conclusion

The source Navier–Stokes construction gives a successful residual-improvement path. The present paper asks whether the repeated path is the smallest known explanation of its success.

The proposed answer is a filtered generator. Same-grade dependence is solved finitely. The strongest linear return appears to gain 0.49996 in filtration. The strongest nonlinear feedback yields a positive difference gain of 0.19998. Finite algebra then reaches any requested residual order, while the source proof’s existing physical realization machinery handles the final infinite assembly.

The larger principle is deliberately compact:

A successful path tells us that a solution can be reached.
A recovered generator tells us which parts of the path ever needed to exist.

A. Notation and role map

Symbol	Role
U_0	Fixed initialized physical state from the source Proposition 9.5.
W	Nonzero angular-harmonic retained residual sector.
$\bar{M}$	Auxiliary-averaged tangential mean residual sector.
M°	Zero-auxiliary-average tangential mean residual sector.
S	Internal compatibility state, including (P, J_θ, J_z) .
$\mathfrak{F}_{\text{adm}}^s$	Admissible filtered augmented-error state of relative grade s .
P_0	One-pass finite same-grade compiler assembled from source primitives.
L_{ret}	Linearization of the retained augmented error about U_0 .
K	Linear return operator $I + L_{\text{ret}}P_0$.
Q_{ret}	Physical quadratic retained error map.
δ_L	Minimum positive linear return degree, proposed as $1/2 - 4\kappa_s$.
η	Nonlinear difference gain $\sigma_0 - 2\kappa_s$.
P_A	Finite Neumann polynomial inverse to target grade A .
X_A	Finite physical correction generated to target grade A .
$\mathcal{I}_{\text{flat}}$	Flat pulse/cutoff remainder ideal, kept outside the compiler state.

B. Exact source crosswalk

Source item	Printed page(s)	Role in the compressed manuscript
Proposition 7.2	77 onward	Supported nonzero-harmonic inverse and oscillatory pressure coefficient.
Proposition 7.6	84 onward	Signed covariance realization $B(W_0, L\Sigma) = \Sigma$; fixed denominator; no sign/relative-size restriction on Σ .
Proposition 8.4 / Corollary 8.5	94–95	Exact integrated compatibility identities and preparation of the compact stress target.
Lemma 8.6	96	Zero-auxiliary-average mean correction.
Lemmas 8.7–8.8	97–99	Fixed five-equation compatibility map and higher-order defect remainder.
Proposition 9.1	101–102	Supported wave residual gain after pulse inversion.
Lemma 9.2	102–103	Wave–mean and wave–wave product rules; quadratic bottleneck.
Proposition 9.3	103–105	Legality of arbitrary finite primitive sequences; pressure reconstruction; retained/flat separation.
Definition 9.4 / Proposition 9.5	106 onward	Stage state and initialized seed state.
Proposition 9.6	107–111	Repeated four-part +0.1 correction cycle; source of the stage architecture being replaced.
Lemma 9.7	111–113	One common domain for every finite construction.
Lemma 9.8	113–114	Physical derivative conversion with losses independent of stage.
Proposition 9.9	114–115	Final realization from finite corrections.
Lemma 5.4	Section 5, esp. 56–61	General shrinking-cutoff summation interface used for final smooth realization.

C. Channel audit matrix

Input	Mechanism	Output	Audit question
W	Proposition 7.2 + Proposition 9.1	W	Does the uncanceled supported residual always land above the target linear grade?
W	covariance interaction	$\bar{M}$	Forward same-grade arrow; is every component passed to stress preparation before the pass ends?
$\bar{M}$	Corollary 8.5 + Proposition 7.6	W	Bottleneck linear return; verify exact typed covariance identity and exponent $1/2 - 4\kappa_s$.
M°	Lemma 8.6	W, S	Verify all backward outputs gain strictly more than δ_L .
S	Lemmas 8.7–8.8	S	Verify exact linear cancellation and higher-order remainder.
$W \times W$	Lemma 9.2	$\bar{M}$	Bottleneck quadratic channel; verify it saturates but does not violate $s + t - 2\kappa_s$.
$W \times M$	Lemma 9.2	W	Verify positive slack.

Input	Mechanism	Output	Audit question
$M \times M$	exact mean equations	M, S	Verify these terms are strictly higher order.
pressure reconstruction	Section 8 exact map	$P, \tilde{M}, M^\circ$	Verify no cubic term is introduced around U_0 .
flat tails	finite allowed operations	flat tails	Verify no allowed finite operator turns flat input into fixed finite order.

D. Authorship, AI assistance, and manuscript status

Author contributions

Michael Zot conceived the MIPU/MAP framework used in this work, developed the Structural Path Compression interpretation pursued here, directed the analysis of the Navier–Stokes construction, selected the mathematical reductions and research questions, and made the final decisions regarding the claims, argument structure, interpretation, and manuscript.

The author is responsible for the mathematical claims and conclusions presented in this manuscript.

AI assistance disclosure

AI systems were used as research tools for literature retrieval, organization of source material, adversarial examination of proof dependencies, symbolic and mathematical checking, exploration of alternative formulations, and editorial drafting. The research direction, conceptual framework, selection and interpretation of mathematical claims, acceptance or rejection of proposed arguments, and final manuscript decisions were determined by the author. AI-generated suggestions were treated as candidate analysis rather than independent mathematical authority and were checked against the cited source construction and the dependency analysis presented here.

Manuscript status

This manuscript presents a proposed stage-free reconstruction of a specific proof architecture in [1]. The central filtration claims are stated with explicit failure criteria to facilitate independent verification. No claim of absolute proof minimality is made; “minimal structure” refers to structure remaining after the representation-induced obligations identified in this analysis are removed.

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